

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

WORK INTEGRATED LEARNING PROGRAMMES

COURSE HANDOUT

Part A: Content Design

Course Title	Mathematical Foundations for Machine Learning
Course No(s)	AMLSIZC416
Credit Units	4
Course Author	Srinath Naidu
Version No	4
Date	
Lead Instructor	Srinath Naidu

Course Description

Vector and matrix algebra, systems of linear algebraic equations and their solutions; Eigenvalues, eigenvectors and diagonalization of matrices; multivariate calculus, vector calculus, Jacobian and Hessian, multivariate Taylor series, gradient descent, unconstrained optimization, constrained optimization, nonlinear optimization, stochastic gradient descent, dimensionality reduction and PCA, optimization for support vector machines.

Course Objectives

No	Objective- The course aims to
CO1	Introduce concepts in linear algebra and to use it as a platform for machine learning.
CO2	Provide techniques for analytical and numerical solutions of linear equations.
CO3	Introduce concepts in multivariate calculus and vector calculus, and introduce multivariate Taylor series.
CO4	Introduce concepts of gradient descent, stochastic gradient descent and the concepts of unconstrained, constrained and nonlinear optimization.
CO5	Introduce the concepts of dimensionality reduction, principal components analysis, and optimization for support vector machines.

Text Book(s)

No	Author(s), Title, Edition, Publishing House
T1	M.P. Dienesroth, A. Aldo Faisal, Cheng Soon Ong, Mathematics for Machine Learning, Cambridge University Press, 2020.
T2	Charu C. Aggarwal, Linear Algebra and Optimization, Springer Nature Switzerland AG, 2020

Reference Book(s) & other resources

No	Author(s), Title, Edition, Publishing House
R1	K Hoffman and R Kunze, Linear Algebra, Pearson Education, 2 nd Edition, 2005.

R2	Erwin Kreyszig, Advanced Engineering Mathematics, Wiley India, 10th Edition, 2015 (earlier editions are also okay)
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Content Structure

No	Title of the module	References
M1	Solution of linear systems – systems of linear equations, matrices, solving systems of linear equations.	T1: Sec 2.1, 2.2, 2.3
M2	Vectors Spaces - linear independence, basis and rank, affine spaces, Norms, inner products, Lengths and distances, Angles and orthogonality, Orthonormal basis	T1: Sec 2.4, 2.5, 2.6, 2.7, 2.8, 3.1, 3.2, 3.3, 3.4, 3.5
M3	Matrix Decomposition methods - Determinant and Trace, Eigenvalues and Eigenvectors, Cholesky decomposition, Eigen-decomposition and diagonalization, singular value decomposition, matrix approximation	T1: Sec 4.1, 4.2, 4.3, 4.4, 4.5, 4.6
M4	Vector Calculus - Differentiation of univariate functions, Partial differentiation and gradients, Gradients of vector-valued functions, Gradients of matrices, Some useful identities for computing gradients, Backpropagation and automatic differentiation	T1: Sec 5.1, 5.2, 5.3, 5.4, 5.5, 5.6, 5.7, 5.8
M5	Continuous Optimization - Optimization using gradient descent, Constrained optimization and Lagrange multipliers, Convex optimization	T1: 7.1, 7.2, 7.3
M6	Nonlinear Optimization - Minutiae of Gradient Descent – learning rate decay, initialization, Properties of optimization in learning – typical objective functions, stochastic gradient descent, how optimization in machine learning is different, tuning hyperparameters, importance of feature pre-processing, Challenges in Gradient-based optimization, local optima and flat regions, differential curvature, examples of difficult topologies like cliffs and valleys, adjusting first-order derivatives for descent, momentum-based learning, AdaGrad, RMSProp, Adam	T2: 4.4, 4.5, 5.2, 5.3
M7	Dimensionality reduction and PCA – problem setting, maximum variance perspective, projection perspective, eigenvector and low-rank approximations, PCA in high dimensions, key steps of PCA in practice, latent variable perspective, Mathematical preliminaries of SVM, primal/dual perspective for SVM, nonlinear SVM - kernels.	T1: 10.1, 10.2, 10.3, 10.4, 10.5, 10.6, 10.7, 12.1, 12.2, 12.3, 12.4, 12.5, 12.6, T2: Sec 6.4 Class Notes

Learning Outcomes:

No	Learning Outcomes
LO1	Students will be able to effectively use matrix algebra tools to analyze and solve systems of linear equations.
LO2	Students will be able to learn concepts of linear algebra that form the foundation of data science problems.

LO3	Students would be able to learn the basic techniques of constrained and unconstrained optimization and understand how these are applied in the context of support vector machines.
LO4	Students will be able to learn the concepts of dimensionality reduction, and principal components analysis.

Part B: Contact Session Plan

Academic Term	Second Semester 2025-2026
Course Title	Mathematical Foundations for Machine Learning
Course No	AIML ZC416
Lead Instructor	Srinath Naidu

Course Contents

Contact Hours	List of Topic Title	Text/Ref Book/external resource
1	Solution of linear systems – systems of linear equations, matrices, solving systems of linear equations	T1: Sec 2.1, 2.2, 2.3
2	Vectors Spaces, linear independence, basis and rank, affine spaces	T1: Sec 2.4, 2.5, 2.6, 2.8
3	Analytic Geometry – norms, inner products, lengths and distances, angles and orthogonality, orthonormal basis	T1: Sec 3.1, 3.2, 3.3, 3.4, 3.5
4	Matrix Decomposition – I Determinant and Trace Eigenvalues and Eigenvectors Cholesky Decomposition	T1: Sec 4.1, 4.2, 4.3

5	Matrix Decomposition – II Eigen-decomposition and Diagonalization Singular Value Decomposition Matrix Approximation	T1: Sec 4.4, 4.5, 4.6
6	Vector Calculus – I Differentiation of univariate functions Partial differentiation and gradients Gradients of vector-valued functions	T1: Sec 5.1, 5.2, 5.3
7	Vector Calculus – II Gradients of matrices Some useful identities for computing gradients Backpropagation and automatic differentiation	T1: Sec 5.4, 5.5, 5.6
8	Vector Calculus – III Higher-order derivatives Linearization and multivariate Taylor's series Computing maxima and minima for unconstrained optimization	T1: 5.7, 5.8 Class Notes
9	Continuous Optimization Optimization using gradient descent Constrained optimization and Lagrange multipliers Convex optimization	T1: Sec 7.1, 7.2, 7.3,
10	Nonlinear Optimization- I Minutiae of Gradient Descent – learning rate decay, initialization Properties of optimization in learning – typical objective functions, stochastic gradient descent, how optimization in machine learning is different, tuning hyperparameters, importance of feature pre-processing	T2: Sec 4.4, Sec 4.5
11	Nonlinear Optimization- II Challenges in Gradient-based optimization, local optima and flat regions, differential curvature, examples of difficult topologies like cliffs and valleys. Adjusting first-order derivatives for descent, Momentum-based learning, AdaGrad, RMSProp, Adam	T2: Sec 5.2, 5.3

12	Dimensionality Reduction and PCA – I Problem setting Maximum variance perspective Projection perspective	T1: Sec 10.1, 10.2, 10.3
13	Dimensionality Reduction and PCA – II Eigenvector Computation and low-rank approximation PCA in high dimensions Key steps of PCA in practice Latent variable perspective	T1: 10.4, 10.5, 10.6, 10.7
14	Mathematical preliminaries for SVM Karash-Kuhn-Tucker conditions	T2: Sec 6.4
15	Primal/dual perspective for linear SVM	T1: 12.1, 12.2, 12.3, 12.4
16	Nonlinear SVM (Expert lecture) kernels examples	Class Notes

Experiential Learning Components:

1. Lab work: 8
2. Project work: 0
3. Case Study: 0
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on the following:

- a) Solving $Ax = b$ using octave and studying the relationship of the accuracy of the solution with the condition number of A.
- b) Gram-Schmidt method of orthogonalization
- c) Studying how eigen-decomposition can be used to compute powers of a matrix.
- d) Calculating eigenvalues using the power method.
- e) Computing singular-value decomposition of a matrix using the eig function in Octave.
- f) Studying the quality of approximation provided by Taylor's series in one and two dimensions, and computing the Hessian of a convex function.
- g) Various flavours of gradient-descent.
- h) Principal components analysis
- i) Linear and non-linear support vector machines (SVM), hard and soft SVM.

Scope of Experiential Learning Component:

Programming language: Python/C/Octave

Tools and libraries: Jupyter, Numpy, Scipy, Pandas, ScikitLearn, Matplotlib, Seaborn, PGMPY

Lab Infrastructure:

Online / Open source / Virtual lab / Google Colab

List of Experiments:

Lab No	Lab Objective	Session Reference
1	Solving $Ax = b$ using Octave and studying the relationship of the accuracy of the solution with the condition number of the matrix.	1,2
2	Gram-Schmidt method of orthogonalization	2,3
3	Finding powers of 3x3 matrix using brute-force matrix multiplication and using EVD(built in eig function in Octave)	4,5
4	Calculation of eigenvalues using the power method	4,5
5	Finding SVD of a 2x3 matrix using built-in Eig function	5
6	Taylor's series in one and two-dimensions, calculating Hessian for a convex function	6,7,8
7	Various flavours of gradient-descent	9,10,11
8	Principal components analysis	12, 13
9	Linear and non-linear support vector machines (SVM), hard and soft SVM	14,15,16

Evaluation Scheme

Legend: EC = Evaluation Component

No	Name	Type	Duration	Weight	Day, Date, Session, Time
1	Assignment 1	Online		10%	To be announced
2	Assignment 2	Online		10%	To be announced
3	Quiz 1	Online	*	5%	To be announced
4	Quiz 2	Online	*	5%	To be announced
5	Mid-Semester Exam	Closed book	90 min	30%	To be announced
6	Comprehensive Exam	Open book	150 min	40%	To be announced

Important Information

Syllabus for Mid-Semester Test (Closed Book): Topics in Weeks 1-8

Syllabus for Comprehensive Exam (Open Book): All topics (in sessions 1 to 16) given in plan of study

Evaluation Guidelines:

1. EC-1 consists of two Assignments and two Quizzes (best two out of the three would be taken for grading). Announcements regarding the same will be made in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
3. For Open Book exams: Use of prescribed and reference text books, in original (not photocopies) is permitted. Class notes/slides as reference material in filed or bound form is permitted. However, loose sheets of paper will not be allowed. Use of calculators is permitted in all exams. Laptops/Mobiles of any kind are not allowed. Exchange of any material is not allowed.
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam. The genuineness of the reason for absence in the Regular Exam shall be assessed prior to giving permission to appear for the Make-up Exam. Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the lectures, and take all the prescribed evaluation components such as Assignment/Quiz, Mid-Semester Test and Comprehensive Exam according to the evaluation scheme provided in the handout.